

Pericardial Effusion & Tamponade — POCUS Quick Reference

PLAX-based identification · differential · tamponade assessment. TeachIM 5-minute POCUS case · CC BY-NC 4.0 · clinician-reviewed.

One line: On the parasternal long axis (PLAX), a **pericardial** effusion lies **anterior** to the descending thoracic aorta; a **pleural** effusion lies **posterior** to it. Most effusions do **not** tamponade — judge by clinical signs, not size.

Identify on PLAX

- **Pericardial** = anterior to descending aorta; tracks *between* heart and aorta.
- **Pleural** = posterior to descending aorta.
- Subxiphoid view = **most sensitive** window for effusion.
- **Always confirm any cardiac POCUS finding in a 2nd view.**
- Note chamber sizes & LV function; circumferential vs loculated.

Differential by Setting

Infectious (more common)
Viral — most common in developed world (Coxsackie, echovirus, HHV6, parvo B19, COVID-19)
TB — >70% in developing world
Non-infectious (less common)
Autoimmune · malignancy · uremia · post-trauma/post-MI · radiation · hypothyroidism

Significance = Size × Chronicity × Clinical

- **Size:** trace may be normal; interpret in context.
- **Chronicity:** compare to prior echo.
- **Tamponade is dynamic** — driven by **rate** of accumulation, not absolute volume (2025 ESC). Small + rapid can tamponade; large + chronic may not.

Tamponade — 3 Key Findings

Sign	Freq %
Elevated JVP	100
Tachycardia	81–100
Pulsus paradoxus >10 mmHg	98
Beck's triad (JVP↑ + ↓BP + muffled tones)	0–19

- Pulsus paradoxus = inspiratory SBP drop >10 mmHg.
- Higher cutoff → ↑specificity, ↓sensitivity.
- **LR+** >10 mmHg: 3.3 · >12 mmHg: 5.9.
- Beck's triad rarely present → do **not** rely on it.

2025 ESC Update

- 2015 ESC pericardial guideline → **2025 ESC myocarditis & pericarditis** guideline.
- **Pericardiocentesis** first-line for significant effusion/ tamponade; surgical window for recurrent/loculated/purulent.
- **CMR** (Class I) if dx unclear/myocarditis suspected; **cardiac CT** (Class I) for loculated/complex effusion, calcification, masses.
- New umbrella concept: **IMPS** (Inflammatory Myopericardial Syndrome).

Bedside Bottom Line

- Confirm location (ant vs post to aorta) in a 2nd view.
- Effusion present? → ask: **JVP↑? tachy? pulsus >10?**
- No tamponade signs → treat the cause (often viral/ supportive).
- Tamponade signs → urgent echo + pericardiocentesis.

Sources: Roy CL, JAMA 2007 (PMID 17456823); Alerhand & Carter, Am J Emerg Med 2019 (PMID 30471929); Stolz L, World J Emerg Med 2017; McGee, Evidence-Based Physical Diagnosis; 2015 & 2025 ESC Guidelines (Eur Heart J). Adapted from TeachIM.org "Pericardial Effusion Without Tamponade – POCUS" (Longino, Fainstad; ed. Cool), CC BY-NC 4.0. Educational use only.